

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning **(BL 5)**

Assessment Tool Weightage: 20%

QUESTIONS: 03

Total Marks:25

Annexure A

Industry Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

<i>Criteria</i>	<i>Conceptual Understanding</i>	<i>Accuracy of Answers</i>	<i>Application of Knowledge</i>	<i>Completeness of Response</i>	<i>Clarity & Presentation</i>	<i>Total</i>
<i>Weightage</i>	25%	25%	20%	15%	15%	
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						

Signature of the student:

Total Marks :

Annexure A

Short Answer Test

1. A banking system wants to detect potential loan defaulters based on customer financial history and transaction patterns. Design a solution using **K-Nearest Neighbour or Logistic Regression**. Explain how similarity or decision boundaries are used, how features are selected, and how the system adapts to new customer data over time.

2. A cybersecurity firm wants to identify abnormal network behavior to detect intrusions. Design a system using **KNN or Case-Based Learning**. Explain how similarity between network patterns is measured, how anomalies are detected, and how the model updates with new attack patterns.

3. An energy management company aims to predict electricity consumption based on factors such as time, weather, and appliance usage. Propose a solution using **Locally Weighted Regression or RBF networks**. Explain how local learning captures variations in consumption patterns and how the model handles noisy and seasonal data.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Machine Learning Solutions for Banking, Cybersecurity, and Energy Management

1. Loan Default Prediction Using K-Nearest Neighbour (KNN)

Proposed Solution

A banking system can use K-Nearest Neighbour (KNN) to classify customers as likely to default or unlikely to default based on their financial history and transaction behaviour.

Features Used

Important features include annual income, loan amount, credit score, debt-to-income ratio, previous loan repayment history, number of delayed payments, account balance, monthly transaction frequency, credit utilization, and employment duration. Numerical features should be normalized or standardized before applying KNN.

Similarity Measurement

KNN determines how similar a new customer is to existing customers. Euclidean distance can be calculated as $d(x,y) = \sqrt{\sum (x_i - y_i)^2}$. The system finds the K most similar historical customers. If most neighbours are defaulters, the new customer is classified as a potential defaulter.

Feature Selection

Features should be selected based on their relationship with loan repayment. Irrelevant or redundant features should be removed using correlation analysis, mutual information, recursive feature elimination, or domain knowledge.

Adapting to New Customer Data

When a new labelled customer record becomes available, it can be added to the training database. The system can periodically revalidate the model as new financial behaviour becomes available.

Advantages

KNN is simple to implement, easy to understand, requires little model training, and can adapt to newly available customer data.

Limitations

KNN can become slow for very large databases, is sensitive to feature scaling, depends on the choice of K, and may be affected by irrelevant features.

2. Intrusion Detection Using KNN / Case-Based Learning

Proposed Solution

A cybersecurity firm can use KNN-based intrusion detection to compare new network traffic with previously observed normal and malicious network patterns. The system can classify activity as normal, suspicious, known attack, or potentially new attack.

Network Features

Useful features include source and destination ports, protocol type, packet size, number of packets, connection duration, bytes transmitted, failed connections, request frequency, TCP flags, and connection rate.

Similarity Measurement

For numerical network features, Euclidean distance can be used: $d(x,y) = \sqrt{\sum (x_i - y_i)^2}$. The system finds the K nearest network patterns and uses majority voting to classify the new event.

Anomaly Detection

The system can compare a new network event with known normal behaviour. If its distance from normal patterns exceeds a selected threshold, the event can be considered anomalous and an alert can be generated.

Case-Based Learning

Previous cybersecurity incidents can be stored as cases containing attack characteristics, network conditions, attack type, response, and outcome. The system follows Retrieve, Reuse, Revise, and Retain: retrieve similar cases, reuse previous knowledge, revise the decision when evidence changes, and retain confirmed new cases.

Updating With New Attack Patterns

When a new attack is confirmed, its characteristics are extracted and added to the case database. This enables future detection of new attacks or variations that resemble known attacks.

Advantages

The approach can recognize patterns similar to previous attacks, incorporate new attack cases, support behavioural anomaly detection, and assist real-time monitoring.

Limitations

Large case databases increase search time. Network features must be carefully selected, attackers can change behaviour, and unusual legitimate traffic may cause false positives.

3. Electricity Consumption Prediction Using Locally Weighted Regression

Proposed Solution

An energy management company can use Locally Weighted Regression (LWR) to predict electricity consumption using time, weather, and appliance usage. LWR gives higher importance to historical observations that are closer to the current prediction point.

Input Features

Possible features include hour of day, day of week, month, temperature, humidity, rainfall, number of occupants, appliance usage, previous

electricity consumption, and holiday/weekend indicators. The target variable is electricity consumption in kWh.

Local Learning

LWR gives higher weights to observations similar to the current situation. A common Gaussian weighting function is $w_i = \exp(-(x - x_i)^2/(2\tau^2))$, where x is the current input, x_i is a historical input, w_i is the weight, and τ is the bandwidth.

Capturing Consumption Variations

Electricity usage differs during mornings, afternoons, evening peaks, weekends, holidays, and different seasons. LWR creates a local model around the current prediction point, allowing it to capture these variations more effectively than a single global linear model.

Handling Noisy Data

Noise may arise from appliance switching, sensor errors, unexpected weather, and temporary interruptions. Noise can be reduced through outlier removal, smoothing, normalization, careful bandwidth selection, and robust preprocessing.

Handling Seasonal Data

Month, season, temperature, and holiday indicators can be included as features. For example, summer consumption may increase because of air-conditioning usage. LWR gives greater importance to historical observations with similar seasonal and weather conditions.

System Architecture

Historical electricity data → Data preprocessing → Feature extraction → Local similarity/weight calculation → Locally Weighted Regression → Electricity consumption prediction → Continuous monitoring and model update.

Advantages

LWR captures local consumption patterns, handles nonlinear variations better than simple global linear regression, and adapts to different weather and usage conditions.

Limitations

It can be computationally expensive for very large datasets, requires appropriate bandwidth selection, is sensitive to poor-quality data, and can become slower as historical data increases.

Overall Comparison

Comparison

Loan default prediction: KNN compares financially similar customers using income, credit score, loan amount, and repayment history. Intrusion detection: KNN/Case-Based Learning compares network behaviour with known patterns using traffic and connection features. Electricity prediction: LWR uses local similarity among time, weather, appliance usage, and previous consumption patterns.

Conclusion

Conclusion

KNN is well suited to loan default prediction because customers with similar financial characteristics may have similar repayment behaviour. The same similarity principle can be applied to cybersecurity, where new network activity is compared with known normal and malicious patterns. For electricity forecasting, Locally Weighted Regression is effective because consumption varies according to time, weather, season, and appliance usage. These approaches can be continuously improved by incorporating new labelled customer records, confirmed attack cases, and recent energy-consumption observations.